

Cross-column RT calibration report

METHOD	cross-column calibration onto a reference column (v2)
REFERENCE COLUMN	Column 25 (bundled default)
STANDARDS PANEL	mix15
STAGE 1 CURVE	595 m/z-matched pairs (173 standards-run + 422 sample-run), 433 kept after outlier trimming
SERIES TERM	series term gated off: leave-own-out MSE 35% above curve-only (threshold: 20% below)
LATTICE TERM	lattice term gated off: leave-own-out MSE 2% above curve-only (threshold: 20% below)
STAGE 2 ANCHORS	not requested (use_sample_anchors=False) - stage 2 is opt-in since 1.2.2
IRT SCALE	1-100, affine on the reference RT axis between the earliest and latest of 14 landmarks (1.46-21.61 min); landmark panel mix15

FEATURES

samples	22,314
standards run	8,768
feature rows in each table	

SERIES + LATTICE

series	off
lattice	off
0 + 0 of 22,314 features corrected	

MATCHED PAIRS

standards	173
sample	422
433 of 595 kept after MAD trimming	

IRT LANDMARKS

14	
RT 1.46-21.61 min on the reference run	

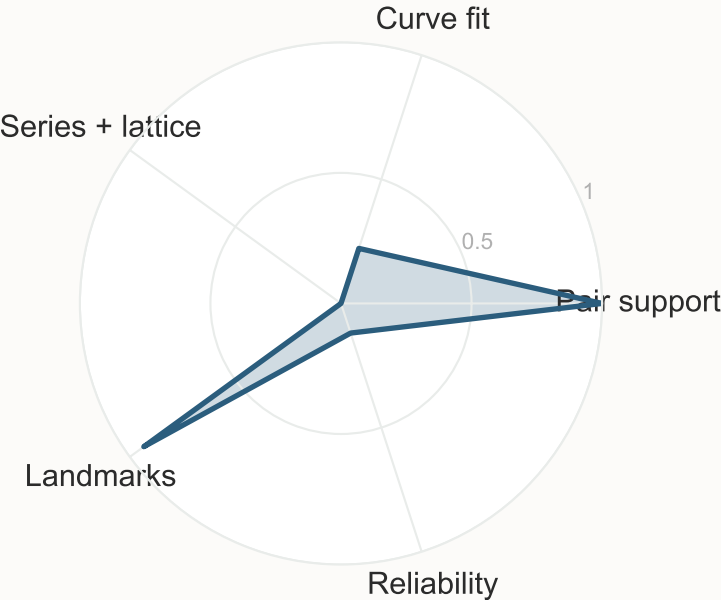
CURVE RESIDUAL

median	0.389
P90	4.245
reference – fitted (min)	

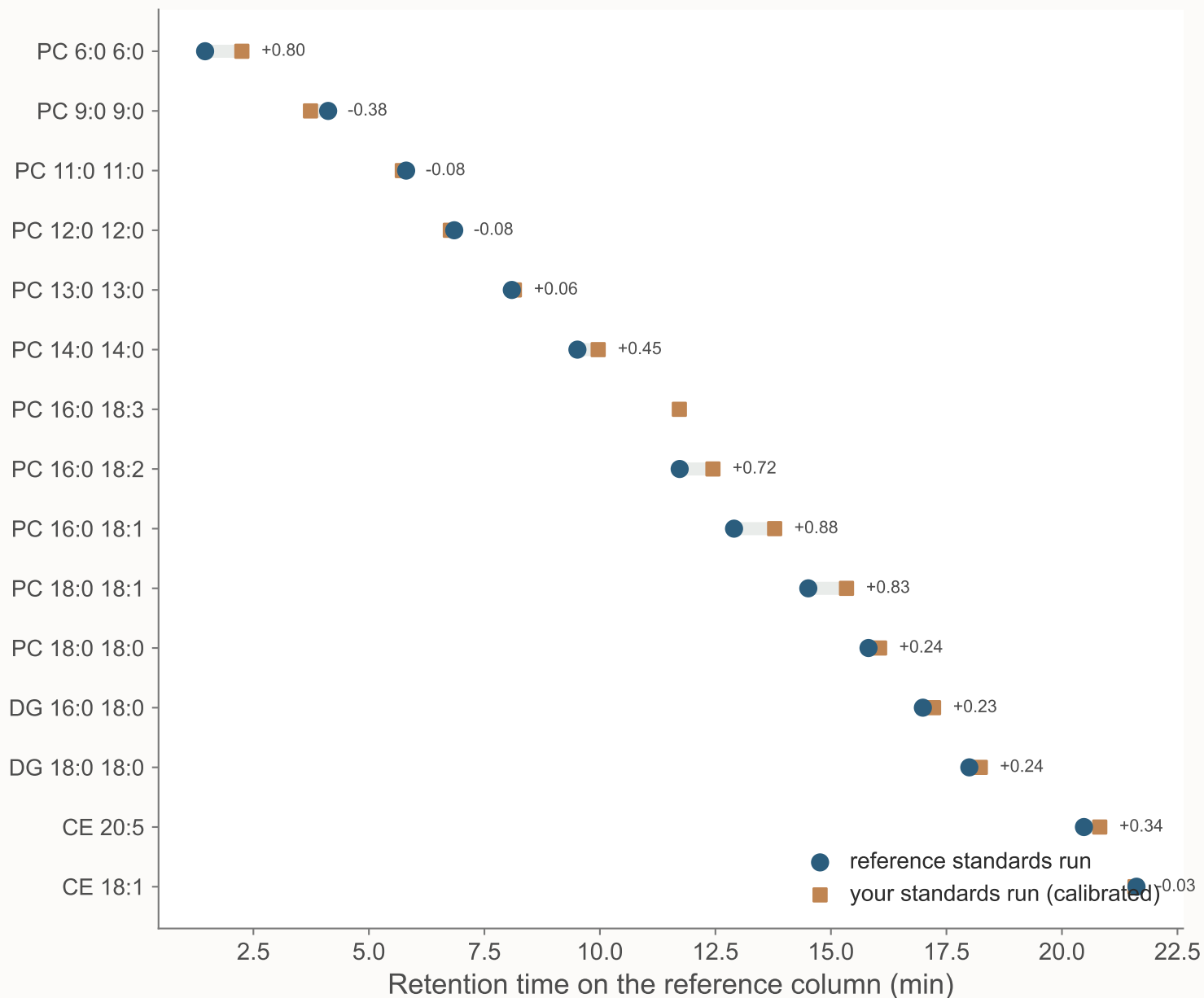
OUTPUT

iRT range	-6-100
extrapolated	2.5%
12% high-reliability	

Quality fingerprint

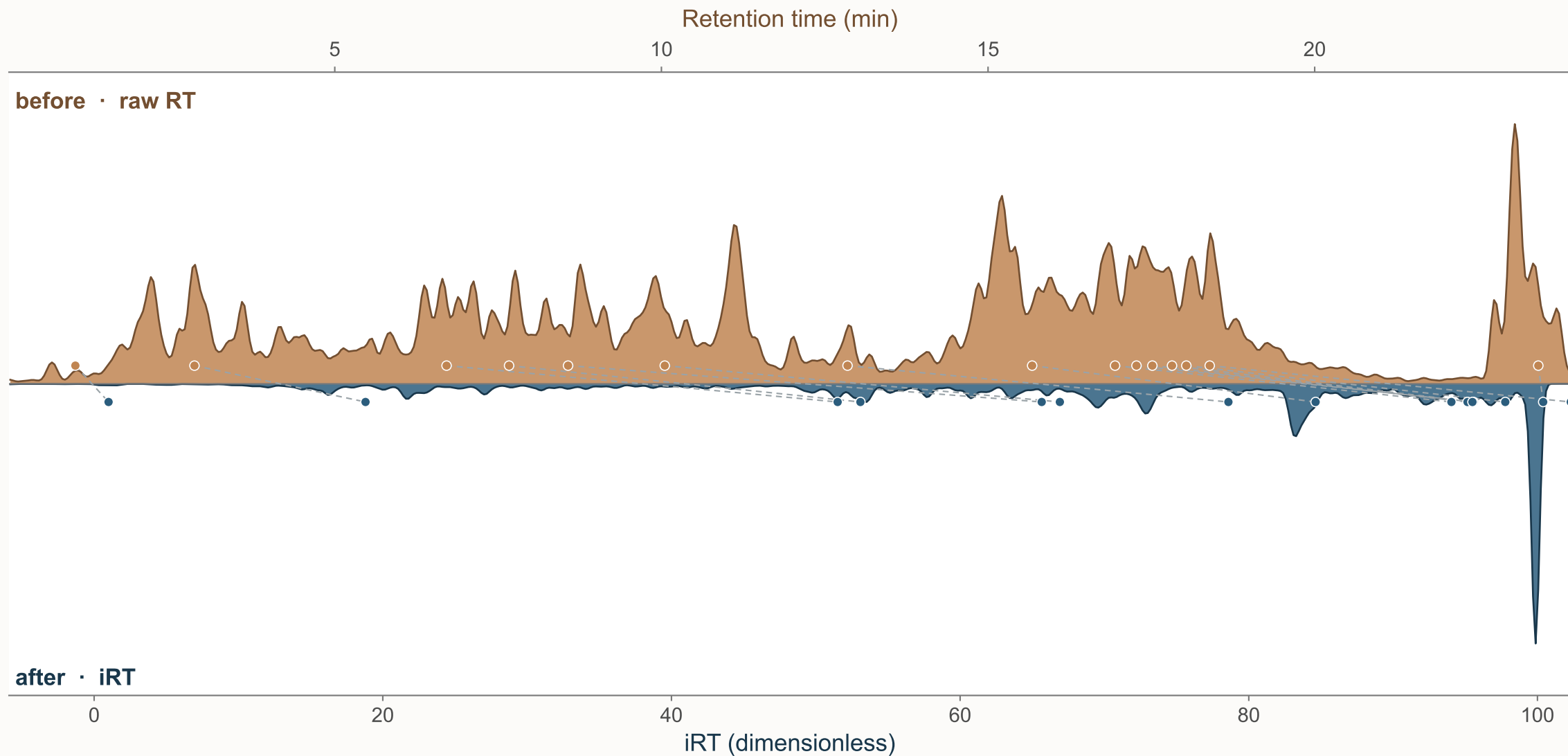


Standard detection — reference vs panel vs samples



Reconstructed feature-intensity profile · before vs after calibration

sampled matched pairs connected — line slant = RT offset · reconstructed from the feature table (binned intensity)



Cross-column calibration curve · your column -> the reference column

series term gated off: leave-own-out MSE 35% above curve-only (threshold: 20% below)

lattice term gated off: leave-own-out MSE 2% above curve-only (threshold: 20% below)

